

A NOTE ON SAATY'S RANDOM INDEXES

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Abstract—The authors present a large, statistically significant set of Critical Indexes for use in Saaty's Analytic Hierarchy Process. In addition, an anomaly in the methods which have been used to calculate critical indexes in the past is pointed out.

Since its development in the 1970s, the Analytic Hierarchy Process [1] (AHP) has achieved widespread popularity as a tool for prioritising the elements in a complex problem in a wide variety of fields, e.g., [2–4].

An important feature of the method is the facility to check the consistency of the decision-maker(s). Several methods for doing this have been suggested [1, 5, 6]. Saaty's preferred method entails the comparison of the Consistency Index (*CI*) of the decision data (based on the right eigenvalue of its matrix) with the Random Index (*RI*), i.e., the *CI* expected from a matrix of that order. This latter is zero for order 2 (i.e., these *must* be consistent) and increases with the order of the matrix.

A list of Saaty's *RI* estimates for positive reciprocal matrices of orders 2 to 14 generated from the integer set 1 to 9 and their enforced reciprocals is provided in [1]. Saaty makes no claims as to the statistical significance of these estimates, carefully warning of the inadequacy of his sample sizes (100 for orders 2 to 11 and 500 for orders 12 to 15). In view of the apparent instability at order 12 and the increased computer power currently available and readily accessible, it is surprising that so little in the way of further statistically acceptable estimates has been published. Liu and Xu's [4] list of *RI*s corresponds largely with Saaty's but supportive details have not been forthcoming. Indeed, the application of AHP to larger groups using a methodology such as that suggested by Donegan *et al.* [7] indicates the need for an extension to the list.

Accordingly, work has been carried out at the University of Ulster to obtain more statistically reliable *RI* estimates.

There are two basic approaches to this type of problem:

- (i) decide on an acceptable tolerance and appropriate confidence limits (e.g., 95%) in the result to determine a significant sample size so that an appropriate random sample can be generated and analysed; or
- (ii) generate a number of large (i.e., over 30) samples for each matrix order so that the *RI* for that order can be estimated as the mean sample mean, with a standard error derived from the sample variances, according to the Central Limit Theorem.

Since the acceptable tolerances would need to vary considerably for the different orders (as can be seen from the Std. Error column in Table 1) ranging from perhaps 0.5 for order 3 to 0.005 for order 20, a certain arbitrariness, in addition to the choice of confidence limits, is involved in method one. The second approach was therefore adopted.

For each matrix order, 10 samples of 100 positive reciprocal matrices were generated from the "set" consisting of the integers 1 to 9 and their reciprocals using a pseudo-random number generator. For historical reasons this was written in Turbo Pascal (C Borland International, Inc.) running on an IBM System/2 Model 70 386 at 20 MHz. The sample size of 100 is (a) large

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and (b) less than 5% of the population in every case: pilot experiments showed no appreciable differences between samples of 100 and 1000. The results of this experiment are shown in Table 1. Estimates for higher order matrices (up to 100) were carried out to enable an investigation of the rate of divergence of the sequence.

Table 1.

Order of Matrix	Saaty Sample Size	Saaty's Avg. <i>CI</i>	Donegan-Dodd Experimental Result		Actual <i>RI</i>
			Avg. <i>CI</i>	Std. Error	
2	100	0.00	0.0000	0.0000	0.0000
3	100	0.58	0.4887	0.0676	0.4914
4	100	0.90	0.8045	0.0609	0.8286
5	100	1.12	1.0591	0.0484	
6	100	1.24	1.1797	0.0389	
7	100	1.32	1.2519	0.0312	
8	100	1.41	1.3171	0.0267	
9	100	1.45	1.3733	0.0235	
10	100	1.49	1.4055	0.0215	
11	100	1.51	1.4213	0.0187	
12	500	1.48	1.4497	0.0165	
13	500	1.56	1.4643	0.0151	
14	500	1.57	1.4822	0.0139	
15	500	1.59	1.4969	0.0127	
16			1.5078	0.0116	
17			1.5153	0.0106	
18			1.5262	0.0102	
19			1.5313	0.00942	
20			1.5371	0.00911	
25			1.5619	0.00723	
30			1.5772	0.00616	
40			1.5976	0.00456	
50			1.6102	0.00357	
60			1.6178	0.00294	
70			1.6237	0.00243	
80			1.6277	0.00215	
90			1.6213	0.00190	
100			1.6339	0.00173	

As a check, exhaustive tests were carried out on reciprocal matrices of order 3 to determine the expected *CI* for (a) entries drawn from the "set" 1 to 9 and their reciprocals (i.e., an aggregate of 18 members), as above, and (b) entries drawn from Saaty numbers 1, 3, 5, 7, 9 and their reciprocals (an aggregate of 10). These yielded results of 0.4914 and 0.4921 respectively. The former can be regarded as a definitive *RI* for matrices of order 3.

A further exhaustive test was carried out using the "set" in (b) above for matrices of order 4. The average *CI* here was 0.8286. In view of the result above, this can be regarded as very close to the exact *RI* for matrices of order 4. A similar test for the 18 member "set" was impractical on the hardware used in view of the time necessary (estimated at 270 hours) to generate and analyse 18^6 matrices.

These values for the low order matrices lend great weight to the *RI* estimates of the author's experiment. The latter, together with their standard errors, indicate that the estimates of Saaty are systematically overlarge. It is unclear why this should be so. One possible explanation is that (at least some of) the matrices in the Saaty experiment were generated using the set of integers from 1 to 9 and their reciprocals, a 17 member set since 1 coincides with its inverse,

1/1 (a practice followed by Budescu *et al.* [6]). This would result in a bias against 1 as a matrix entry. However, since all possible matrix entry candidates must be equally likely, the probability of an entry being 1 must be twice that of the other numbers if the concept of *expected CI* is to have any meaning and this bias would be incorrect. Our limited experiments on these lines proved inconclusive: the average *CI*s for matrices randomly generated using the 17 member sets tended to be about halfway between the experimental and Saaty's estimates. On the other hand, the *CI* values for matrices generated without any units other than on the leading diagonals were about the same amount less than the experimental results.

Also, since the *RI* depends on the generating set used [5], it might make sense to standardise the raw data to a range inside 1 to 9 and their inverses. For example, Budescu's 49 point set [6] would be reduced to 1, 1.33, 1.67, 2, ..., 9 and their inverses. This would allow our set of *RI*s to be used in all consistency checks since the differences between the *RI*s for 1 to 9 and 1, 3, 5, 7, 9 do not appear to be very large.

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